WAJID HUSSAIN

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RESEARCH INTERESTS

Energy Systems and Intelligent Control, Data-Driven Optimisation of Energy Systems, Cyber-Physical Systems and IoT Sensing, Applied Machine Learning for Energy Systems.

EDUCATION

Master of Science in Cyber-Physical Systems (IoT) Sep 2023 to Apr 2025
Northeastern University, Toronto, ON, Canada
GPA: 3.798/4.00

Graduate Certificate in Business Administration (Online) Sep 2025 to Apr 2026
Northeastern University, D'Amore-McKim School of Business

Bachelor of Technology in Electrical and Electronics Engineering Aug 2004 to Apr 2008
Jawaharlal Nehru Technological University (JNTU), Hyderabad, India
Marks: 64.32% (First Class)

RESEARCH AND TECHNICAL PROJECTS

Intelligent Solar Panel Orientation System (ISPOS) (wajidengg.github.io/ispos) Jan 2025 to Apr 2025
Graduate Capstone Project, Northeastern University
Advisors: Dr. Rolando Herrero, Dr. Haitham Tayyar

- Designed a system that decides how to point a solar panel by combining three inputs: sun detection from an ESP32-CAM camera, sun-position calculations (PVLib) and live weather data (Open-Meteo), so the system adjusts to clear or cloudy skies.
- Built a 10-year hourly dataset for Toronto (~87,700 records) combining sun position and weather, and used it to study how seasons and cloud cover affect solar potential.
- Built the monitoring setup with ESP32, LoRa, MQTT and a ThingsBoard dashboard on AWS EC2, and compared four sun-position libraries (PVLib, Skyfield, Ephem, PySolar) for accuracy.

MQTT-Based Industrial Telemetry and Real-Time Dashboard
Independent Technical Project, 2026

- Built a data pipeline that simulates temperature readings from several production lines, sending them through a Mosquitto MQTT broker using Python, with organised topics for each line.
- Created a live Node-RED dashboard to display the data, running Mosquitto and Node-RED in separate Docker containers.

ACADEMIC EXPERIENCE

Graduate Teaching Assistant Jan 2024 to Apr 2025
Northeastern University, Toronto, Canada

- Supported delivery of Fundamentals of IoT (TELE6510) across three semesters.
- Held office hours, evaluated assignments and guided students.

PROFESSIONAL ENGINEERING EXPERIENCE

Electrical Engineer Sep 2014 to Dec 2020

ABV Rock Group Ltd., Riyadh, Saudi Arabia

- Coordinated installation, testing, commissioning and handover of electrical systems including BMS, CCTV and access control, at KSU Medical City and Security Forces Medical City projects.
- At KSU Medical City, also coordinated installation, testing, commissioning and handover of 50+ elevators and five 2.5 MVA standby generators with synchronizing panels.
- Completed snag-list and handover of electrical systems for a Ministry of Interior HQ secure helipad facility and the King Salman Center at Prince Sultan University.
- Led teams of 20 to 105 technicians on site and coordinated with clients, consultants, suppliers, design, QC and procurement teams to resolve technical issues and keep site execution on schedule.

Electrical Engineer Jan 2013 to Apr 2014

Dubai Contracting Company, Riyadh, Saudi Arabia

- Reviewed drawings and coordinated electrical installation, materials and subcontractors for Burj Rafal Kempinski (the tallest tower in Riyadh at completion) and Ascott Rafal.
- Resolved design conflicts between consultant and contractor teams before installation and carried out quality inspections during site execution. Coordinated testing, commissioning and final handover.

Electrical Engineer May 2010 to Jul 2012

World Energy Contracting Company, Dammam, Saudi Arabia

- Supervised fire alarm installation at Aramco North Riyadh and MEP installation at Merck & Co. Riyadh.
- Led installation, testing and commissioning of digital fault recorders at Saudi Electricity substations.

ADDITIONAL EXPERIENCE

Sales Supervisor Jul 2025 to Present

Freedom Mobile Authorized Dealer, Courtenay, BC, Canada

- Supervised daily store operations, staff performance, inventory and customer service.
- Achieved the highest store target attainment in Canada for three consecutive months.

TECHNICAL SKILLS

- **IoT Systems:** ESP32/ESP32-CAM, ESP-IDF, LoRa, MQTT (Paho, Mosquitto), CoAP, ThingsBoard.
- **Programming and Tools:** Python, Java, Git/GitHub, Docker, Linux, AWS.
- **Building Services and MEP:** AutoCAD, MEP coordination, electrical and low-current systems (BMS, CCTV, access control), site inspection and handover.

PROFESSIONAL REGISTRATION AND MEMBERSHIPS

Chartered Engineer (C.Eng) and Member (MIE), Institution of Engineers (India), 2017.
Saudi Council of Engineers, registered 2013 to 2021.